



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Modeling and Stimulation

Lab Experiment 2

MTECH-COMPUTER SCIENCE
ENGINEERING
CYBER SECURITY AND FORENSICS

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Aim: To implement and compare discrete and continuous population growth models, analyze their differences through simulation, and evaluate their advantages, limitations, and real-world applicability using mathematical and numerical approaches.

Theory :

Mathematical modelling is widely used to represent real-world systems. Population growth is one such system that can be modeled using either discrete or continuous approaches.

1. Discrete Modelling

In discrete modelling, population changes occur at fixed intervals (e.g., yearly). The population at the next step depends only on the previous step.

The discrete exponential growth equation is:

$$P_{n+1} = P_n (1 + r)$$

where:

- P_n = population at time step n
 r = growth rate

This model is suitable for systems where updates occur periodically.

2. Continuous Modeling

In continuous modelling, population changes smoothly over time and is represented by a differential equation:

$$\frac{dP}{dt} = rP$$

Solving this equation gives:

$$P(t) = P_0 e^{rt}$$

where:

- P_0 = initial population

- r = growth rate
 t = time

This produces smooth exponential growth.

3. Logistic Growth (Realistic Extension)

Exponential growth assumes unlimited resources, which is unrealistic. A more practical model is logistic growth:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$$

Where: K = carrying capacity

This model limits population growth as it approaches environmental constraints.

Methodology:

2.1 Parameters Used

- Initial Population $P_0 = 100$
 Growth Rate $r = 0.1$
 Carrying Capacity $K = 1000$
 Maximum Time $t_{max} = 20$
- Time Step (Euler method) $dt = 0.1$

2.2 Models Implemented

1. Discrete exponential model
2. Continuous analytical exponential model
 Continuous numerical model using Euler's method
3. Logistic growth model
4. Percentage error analysis between discrete and continuous models

Python libraries used:

- NumPy (numerical computation)
- Matplotlib (visualization)

Code :

```
lab2.py > ...
1  # Author : Jigesh Sheoran
2  # Last Modified : 12/02/2026
3
4  import numpy as np
5  import matplotlib.pyplot as plt
6
7  # PARAMETERS
8  P0 = 100
9  r = 0.1
10 K = 1000          # carrying capacity
11 t_max = 20
12 dt = 0.1          # small timestep for numerical solution
13
14 # DISCRETE EXPONENTIAL MODEL
15 time_discrete = np.arange(0, t_max + 1, 1)
16 population_discrete = [P0]
17
18 for i in range(1, len(time_discrete)):
19     P_next = population_discrete[i - 1] * (1 + r)
20     population_discrete.append(P_next)
21
22 population_discrete = np.array(population_discrete)
23
24 # CONTINUOUS ANALYTICAL EXPONENTIAL
25 time_cont = np.linspace(0, t_max, 200)
26 population_cont = P0 * np.exp(r * time_cont)
27
28 # CONTINUOUS NUMERICAL (Euler Method)
29 time_euler = np.arange(0, t_max, dt)
30 population_euler = [P0]
31
32 for t in time_euler[:-1]:
33     P_current = population_euler[-1]
34     dP = r * P_current
35     population_euler.append(P_current + dP * dt)
36
37 population_euler = np.array(population_euler)
38
39 # LOGISTIC MODEL (Continuous)
40 population_logistic = [P0]
41
42 for t in time_euler[:-1]:
43     P_current = population_logistic[-1]
44     dP = r * P_current * (1 - P_current / K)
45     population_logistic.append(P_current + dP * dt)
46
47 population_logistic = np.array(population_logistic)
48
49 # ERROR ANALYSIS
50 interp_cont = np.interp(time_discrete, time_cont, population_cont)
51 error = np.abs(population_discrete - interp_cont)
52 percent_error = (error / interp_cont) * 100
53
```

```

53
54 # PLOTS
55 plt.figure(figsize=(10, 6))
56 plt.plot(time_discrete, population_discrete, 'o-', label='Discrete Model')
57 plt.plot(time_cont, population_cont, '--', label='Continuous Analytical')
58 plt.plot(time_euler, population_euler, ':', label='Continuous (Euler)')
59 plt.plot(time_euler, population_logistic, label='Logistic Model')
60
61 plt.xlabel("Time")
62 plt.ylabel("Population")
63 plt.title("Discrete vs Continuous Population Growth Models")
64 plt.legend()
65 plt.grid(True)
66 plt.show()
67
68 # ERROR GRAPH
69 plt.figure(figsize=(8, 5))
70 plt.plot(time_discrete, percent_error, 'r-o')
71 plt.xlabel("Time")
72 plt.ylabel("Percentage Error (%)")
73 plt.title("Percentage Error: Discrete vs Continuous")
74 plt.grid(True)
75 plt.show()
76
77 # PRINT METRICS
78 print("Final Population Values at t =", t_max)
79 print("Discrete:", population_discrete[-1])
80 print("Continuous Analytical:", population_cont[-1])
81 print("Euler Numerical:", population_euler[-1])
82 print("Logistic Model:", population_logistic[-1])
83 print("\nMaximum Percentage Difference:", max(percent_error), "%")
84

```

Terminal Output :

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh + v

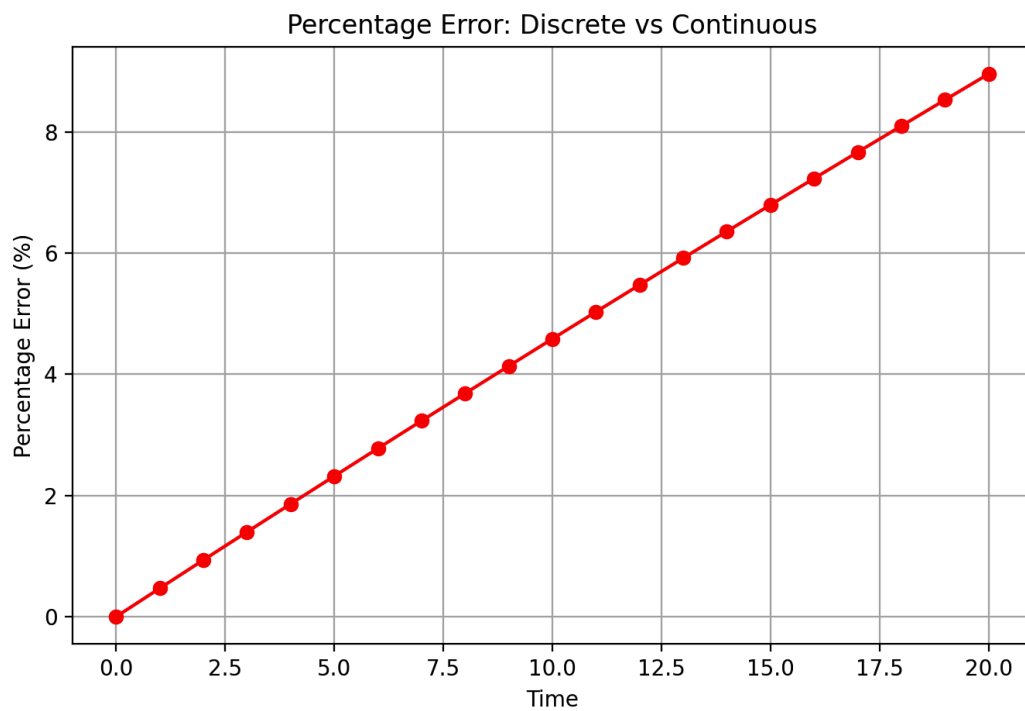
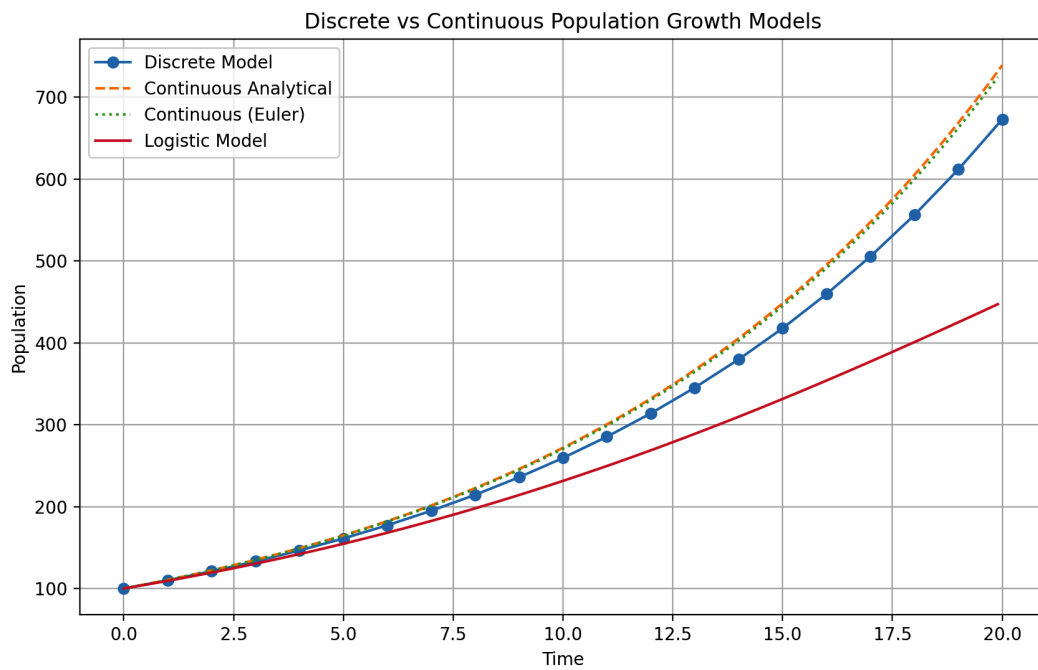
```

/usr/local/bin/python3 "/Users/sheoraninfosec/Documents/Documents - Jigesh's MacBook Air/Masters eling and Stimulation/Lab 2/lab2.py"
● sheoraninfosec@Jigeshs-MacBook-Air Lab 2 % /usr/local/bin/python3 "/Users/sheoraninfosec/Documen h's MacBook Air/Masters UPES/Semester 2/Modeling and Stimulation/Lab 2/lab2.py"
Final Population Values at t = 20
Discrete: 672.7499949325611
Continuous Analytical: 738.905609893065
Euler Numerical: 724.3582031514804
Logistic Model: 447.1290348040087

Maximum Percentage Difference: 8.953188888372095 %
○ sheoraninfosec@Jigeshs-MacBook-Air Lab 2 % 

```

Graphical Output :



Results :

Model	Final Population
Discrete Model	672.75
Continuous Analytical	738.91
Continuous (Euler)	724.36
Logistic Model	447.13

Growth Curve Behavior

The simulation produced four distinct curves:

1. The **Discrete Model** shows stepwise exponential growth and remains slightly lower than the continuous analytical curve due to periodic updating.
2. The **Continuous Analytical Model** exhibits smooth and uninterrupted exponential growth, resulting in the highest population value at $t = 20$.
3. The **Euler Numerical Method** closely follows the analytical solution with minor deviation caused by numerical approximation.
4. The **Logistic Model** initially grows exponentially but gradually slows and stabilizes due to the carrying capacity constraint, representing bounded and realistic growth.

Error Analysis

From the percentage error graph:

- Error increases almost linearly over time.
- Maximum percentage difference observed: 8.95%
- Error accumulation occurs because discrete compounding differs from continuous exponential growth.

This confirms that discrete approximation diverges gradually from continuous behaviour over longer durations.

COMPARATIVE MATRIX TABLE

Feature	Discrete Model	Continuous Analytical	Continuous (Euler)	Logistic Model
Nature of Change	Stepwise	Smooth	Smooth (approx.)	Smooth & bounded
Mathematical Basis	Difference Equation	Differential Equation	Numerical ODE	Differential Equation
Realism	Moderate	Idealized	Approximation	High
Computational Complexity	Low	Low	Moderate	Moderate
Long-term Stability	Unbounded	Unbounded	Unbounded	Bounded
Error Accumulation	Yes	No	Depends on dt	Minimal

Conclusion

The experiment demonstrates that discrete and continuous models yield similar short-term behaviour but diverge progressively over longer time periods. At $t = 20$, the discrete model underestimates the population by approximately 8.95% compared to the continuous analytical solution.

The continuous analytical model produces the highest population due to uninterrupted exponential growth, while the Euler method closely approximates this behaviour with minor numerical deviation. In contrast, the logistic model shows significantly lower growth due to carrying capacity constraints, making it the most realistic representation of natural population systems.

The results confirm that model selection depends on system characteristics: discrete models are suitable for periodic updates, continuous models for theoretical analysis, and logistic models for realistic bounded growth scenarios.

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